

Multiple Choice (10 marks)

1. Which of the following statements about a remote procedure call is true?
 - (a) It is used to call procedures with addresses that are farther than 2^{16} bytes away.
 - (b) It cannot return a value.
 - (c) It cannot pass parameters by reference.
 - (d) It cannot call procedures implemented in a different language.
2. A sender must not be able to deny sending a message that was sent, is known as
 - (a) Message Nonrepudiation
 - (b) Message Integrity
 - (c) Message Confidentiality
 - (d) Message Sending
3. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port
4. Which of the following comes closest to being a perfectly secure encryption scheme?
 - (a) The Caesar Cipher, a substitution cipher
 - (b) DES (Data Encryption Standard), a symmetric-key algorithm
 - (c) Enigma, a transposition cipher
 - (d) One-time pad
5. The language $\{ww \mid w \text{ in } (0 + 1)^*\}$ is
 - (a) not accepted by any Turing machine
 - (b) accepted by some Turing machine, but by no pushdown automaton
 - (c) accepted by some pushdown automaton, but not context-free
 - (d) context-free, but not regular

True / False (10 marks)

Answer True or False

6. True or False: Authorization aims to restrict what operations and data a user can access within a system.
7. True or False: The four primary security principles related to messages are Confidentiality, Integrity, Non-repudiation, and Authentication.
8. True or False: A buffer is a sequential segment of memory allocated for storing data such as character strings or arrays of integers.
9. True or False: Message confidentiality ensures that only the intended sender and receiver can access the content of the communication.
10. True or False: Bubble sort has average-case and worst-case running times of $O(n \log n)$, which makes it suitable for large datasets.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the minimum product from the pairs of tuples within a given list.
12. Write a python function to count the number of lists in a given number of lists.

End of Paper